جـامـعـة الـشـرق الأوسط
MIDDLE EAST UNIVERSITY
Amman - Jordan

Course Syllabus

General Information

Faculty	Information Technology		
Department	Software Engineering		
Semester	1st	**Academic Year**	2025-2026

Course Details

Course Title	Software Requirements Engineering			**Course No.**	0442501		
Type of Learning	■ On- campus						
Credit Hours	3	**Theoretical**	3	**Practical**	0		
Course Level (according to JNQF)			7	**JNQF Hours***	12.15		
Pre-requisite	Software Engineering Fundamentals-0442401			**Co-requisite**			

Course Instructor Information

Instructor Name	Prof. Qusai Shambour		
Instructor E-mail	q.shambour@meu.edu.jo		
Course Time	(11:00 – 12:00 Sun., Tues., and Thur.)		
Office Hours	Sun., Tues., and Thur. (14:00 - 16:00)		
Office Number	N	**Office Phone**	300
Lab Instructor Name –If assigned-			

Sources and References

1. Textbook

Wiegers, K., & Hokanson, C. (2023). Software Requirements Essentials: Core Practices for Successful Business Analysis, 1st Edition, Pearson.

2. References

1) Robertson, S., Robertson, J. and Reed, A. (2024), Mastering the Requirements Process, 4th Edition, Pearson.
2) Pohl, K. and Rupp, C. (2015), Requirements Engineering Fundamentals, (2nd Edition), Rocky Nook Inc.

Teaching Methods

1. Interactive Lectures
2. Discussion
3. Guest Lectures
4. Case-Based Learning
5. Problem-Based Learning
6. Collaborative Group Projects

Course Description

This course introduces the principles, processes, and practices of software requirements engineering as the foundation of successful software development. The course emphasizes understanding business needs, defining clear objectives, and transforming them into precise, verifiable software requirements. Students will learn to elicit, analyze, specify, validate, and manage software requirements through an iterative and collaborative approach. Topics include identifying stakeholders and decision makers, defining solution boundaries, modeling functional and nonfunctional requirements, developing use cases, assessing data and quality attributes, prioritizing requirements, managing requirement baselines and changes, and validating deliverables against business objectives.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLOs	PLOs	JNQF Descriptors		
			Knowledge	Skill	Competency
1	Explain the role and significance of requirements engineering, detailing its fundamental activities and the classification of software requirements (e.g., functional, non-functional, and quality attributes).	PLO1	K1–I		
2	Explain appropriate elicitation techniques to identify and document stakeholder requirements.	PLO1	K1–I		
3	Analyze and prioritize gathered requirements to ensure clarity, completeness, and feasibility.	PLO1	K1–P		
4	Develop and document a structured Software Requirements Specification (SRS) using standard modeling and documentation techniques.	PLO1	K1–P		
5	Explain validation and management processes to ensure software requirements are correct, complete, and aligned with stakeholder expectations.	PLO1	K1–P		
6	Communicate software requirements effectively to diverse audiences (e.g., team members, clients) using multiple professional formats, such as written specifications, diagrams, and oral presentations.	PLO3			C1–P

Course Topics

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture No.	Lecture Type		Assessment Tool	Summative/ Formative
1	**Topic 1: Introduction and Foundations of Requirements Engineering** • Definition and scope of Requirements Engineering • Importance and impact on project success • Core activities and lifecycle stages	Lecture (1)	On Campus	1	Discussion & Questions	Formative
		Lecture (2)	On Campus	1	Discussion & Questions	Formative
		Lecture (3)	On Campus	1	Discussion & Questions	Formative
2-4	**Topic 2: Types and Characteristics of Requirements** • Functional, Non-functional, and Quality	Lecture (1)	On Campus	1	Discussion & Questions	Formative
		Lecture (2)	On Campus	1	Discussion & Questions	Formative

	attributes • Characteristics of good requirements • Skills of a requirements engineer	Lecture (3)	On Campus	1	Discussion & Questions	Formative
5	**Topic 3: Context, Boundaries, and Stakeholders** • System and environment boundaries • Identifying stakeholders and decision makers • Understanding business context and goals	Lecture (1)	On Campus	2	Discussion & Questions	Formative
		Lecture (2)	On Campus	2	Discussion & Questions	Formative
		Lecture (3)	On Campus	2	Project Demonstration	Formative
6	**Topic 4: Requirements Elicitation** • Requirement sources • Elicitation Techniques	Lecture (1)	On Campus	2	Discussion & Questions	Formative
		Lecture (2)	On Campus	2	Discussion & Questions	Formative
		Lecture (3)	On Campus	2	Discussion & Questions	Formative
7-9	**Topic 5: Requirements Analysis and Prioritization** • Analyzing requirements for completeness and consistency • Detecting conflicts and dependencies • Prioritization methods	Lecture (1)	On Campus	3	Discussion & Questions	Formative
		Lecture (2)	On Campus	3	Discussion & Questions	Formative
		Lecture (3)	On Campus	3	Discussion & Questions	Formative
8	Mid Exam					
10-11	**Topic 6: Requirements Specification and Documentation** • SRS structure and IEEE standards • Use cases and modeling techniques (UML, user stories) • Writing clear, consistent, and testable requirements	Lecture (1)	On Campus	4	Project Phase 1	Formative
		Lecture (2)	On Campus	4	Discussion & Questions	Formative
		Lecture (3)	On Campus	4	Discussion & Questions	Formative
12	**Topic 7: Requirements Validation and Verification** • Quality criteria for requirements • Validation techniques (reviews, inspections, prototyping) • Aligning requirements with stakeholder expectations	Lecture (1)	On Campus	5	Discussion & Questions	Formative
		Lecture (2)	On Campus	5	Discussion & Questions	Formative
		Lecture (3)	On Campus	5	Project Phase 2	Formative
13	**Topic 8: Requirements Management and Change Control** • Traceability and versioning • Managing evolving requirements • Change control and configuration management	Lecture (1)	On Campus	5	Discussion & Questions	Formative
		Lecture (2)	On Campus	5	Discussion & Questions	Formative
		Lecture (3)	On Campus	5	Discussion & Questions	Formative
14-15	**Project**	Presentation		1-6	Presentations and Discussions	Summative
16	**Final Exam**	Assessment		1-6	Questions	Summative

Course Evaluation Time and Weight

Assessment Tool	Weight %	Expected Due Date
Quiz	-	
Project	35%	Week 14
Mid Exam	25%	Week 8
Final Exam	40%	Week 16

Course Policies

Item	Policy
Quizzes	- No makeup quizzes. - A minimum 2 quizzes are given. - Each Quiz is out of 5. - Failure in attending a quiz will result in zero mark unless the student provides an official acceptable excuse to the instructor who approves a makeup quiz.
Exams	- The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyse a Problem, Essay Questions, etc.
Makeup Exams	- Failure in attending a course exam other than the final exam will result in zero mark unless the student provides an official acceptable excuse to the instructor who approves a makeup exam. - Failure in attending the final exam will result in zero mark unless the student presents an official acceptable excuse to the Dean of his/her faculty who approves an incomplete exam, normally scheduled to be conducted during the first two weeks of the successive semester.
Drop Date	- Last day to drop the course is according to the University regulations which is 15/06/2024
Academic Honesty	- Cheating or copying on exam or quiz is an illegal and unethical activity. - Standard MEU policy will be applied. - All graded assignments must be your own work (your own words). - An attempt of an individual to claim the work of another as the product of his/her own thoughts, regardless of whether that work has been published. - Quoting improperly or paraphrasing text or other written materials without proper citation on an exam, term paper, homework, or other written material submitted to an Instructor as one's own work - Handing in a paper to an instructor that was purchased from a term paper service or downloaded from the Internet and presenting another person's academic work as one's own.
Attendance	- Excellent attendance is expected. - MEU policy requires the faculty member to assign ZERO grade if a student misses 10% of the classes. - Sign-in sheets will be circulated. - If you miss class, it is your responsibility to find out about any announcements or assignments you may have missed.
E-Learning	- It is your responsibility to check the course’s eLearning website regularly (at least once every 2-3 days) for announcements and material.
Workload	- The student should expect to spend 6 hours per week as an average workload.
Graded Exams	- Instructor should return exam papers graded to students within one week after the exam date.
Participation	- Participation in and contribution to class discussions will affect your final grade positively. - Raise your hand if you have any question. - Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	- Assignments and projects should be submitted to the instructor on the due date. Zero mark will be given for late submissions unless the student has an acceptable excuse approved by the instructor of the course. - Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application. - More details about phases and topics of the projects will be given later on.
Others	- All mobile phones and/or other communication devices should be turned off before entering the classroom.

Rubrics and Feedbacks (If Applicable)

Software Requirements Specification (SRS) Project Evaluation Rubric

This rubric is designed to evaluate the Software Requirements Specification (SRS) project for the course Software Requirements Engineering. It assesses students' abilities in applying requirements engineering concepts, elicitation, analysis, specification, validation, management, and communication, in line with the course CLOs and PLOs.

Criterion	Weight (%)	Excellent (90–100)	Good (75–89)	Satisfactory (60–74)	Needs Improvement (<60)	Mapped CLOs
Problem Understanding & Context Definition	10%	Demonstrates deep understanding of the business problem, context, and boundaries; identifies all key stakeholders and system interfaces.	Defines problem and context clearly with minor omissions; identifies most stakeholders.	Demonstrates partial understanding of problem and context; limited stakeholder analysis.	Shows poor understanding of problem scope; key elements missing.	CLO1, CLO2
Elicitation Process & Requirement Gathering	10%	Applies multiple elicitation techniques effectively; findings are well-documented and traceable.	Uses at least two elicitation methods with adequate documentation.	Uses limited elicitation techniques; documentation lacks depth.	Minimal or unclear elicitation evidence; weak stakeholder engagement.	CLO2
Requirement Analysis & Prioritization	20%	Thoroughly analyzes requirements for clarity, completeness, and feasibility; uses formal prioritization methods.	Performs sufficient analysis with logical prioritization and clear rationale.	Provides basic analysis but with incomplete prioritization.	Analysis is superficial or lacks justification.	CLO3
Requirements Specification (SRS Content & Structure)	20%	Produces a professional, complete SRS following IEEE standards; requirements are clear, consistent, and verifiable.	SRS mostly complete and well-organized with minor inconsistencies.	SRS includes essential components but lacks clarity or structure.	SRS incomplete or poorly organized; lacks essential sections.	CLO4
Modeling & Diagrams (Use Cases)	15%	Models are complete, correct, consistent, and enhance understanding of	Models are mostly accurate and readable with minor inconsistencies.	Models are partially correct but limited or inconsistent with SRS.	Models are inaccurate or missing.	CLO4

		requirements.				
Validation, Verification & Change Management	10%	Demonstrates systematic validation and verification; maintains full traceability and clear change management plan.	Includes validation and verification with some evidence of review.	Minimal validation efforts; lacks structured change control.	No validation process or traceability evidence.	CLO5
Professional Documentation & Presentation Quality	15%	Writing is precise, clear, and professionally formatted; excellent visual organization, references, and glossary included.	Documentation clear with minor formatting issues.	Acceptable structure but lacks professional formatting.	Unclear writing, poor formatting, and missing standards.	CLO4, CLO6
Total Weight	**100%**					

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			
Lectures and Seminars	3	15	45
Tutorials	–	–	–
laboratory	–	–	–
Activities for Assessment and Evaluation			
First Exam/ Midterm	1	1	1
Second Exam (If applicable)	–	–	–
Final Exam	2	1	2
Quizzes	–	–	–
Self-Learning Activities			
Assignments/ Project	4	8	32
Case Studies	–	–	–
Presentations	–	–	–
E-learning	–	–	–
Extra Readings	3	14	42
Total Hours	**122**		
JNQF Hours	**Total Hours/10 = 12.2**		

*** JNQF Hours Calculation**